1. Lead R5.1: the quotient-coordinate oracle in characteristic 2

Round 5, 2026-10-12. Example `examples/r5_1_quotient_oracle.rs`. Raw records in this directory: `validate.json`, `grid_small.json`, `grid_n23.json`, `grid_n31.json`, `grid_n37_m23.json`, `grid_n37_m4.json`, `a1_m31.json`, and the Rust-computed pooled statistics `summary.json`. Full tables, commands and limits are in `RESULTS.md`. Toy Koblitz curves only. No sub-rho claim and no claim about deployed curves. Wall times are contended.

1.1. Question

On $y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, the 2-torsion kernel $M = \langle T \rangle$ has $T = (0, 1)$. Prop. 15.2 charges a $+T$-saturated base 2^{m-2} for the mitm oracle. Round 4 (§17.4, $\ell = 3$, odd characteristic) found this charge only for oracles that enumerate F. Hypothesis: a quotient oracle (arm c) costs about 1 at $m = 3$, and (c) ≈ (d) up to the index factor 2. Falsifier: (c) still pays about 2^{m-2}.

1.2. The 2-isogeny (derived; validated)

Vélu for the 2-torsion point $(0, s)$, $s = \sqrt{b}$, gives $\varphi(x, y) = (x + s/x, y + s(x + y + s)/x^2 + s)$ onto $y^2 + xy = x^3 + ax^2 + s$. For $b = 1$ the codomain is E itself, $\varphi = \pm(\pi \mp 1)$ is the Verschiebung, and the quotient coordinate is $x' = x + 1/x$.

Measured on 15 curves (5 Koblitz curves and 10 random $b \notin \mathbb{F}_2$; $n \leq 17$):

- 0 points off the codomain;
- 4000/4000 homomorphism checks;
- kernel exactly $\{O, T\}$;
- $\#E' = \#E$;
- image index exactly 2, equal to $\{Q : [N/2]Q = O\}$.

The dual is the Frobenius, which is inseparable, so $\hat{\varphi}(E') = E$. The index-2 image control is therefore $2E = \varphi(E)$.

1.3. Results

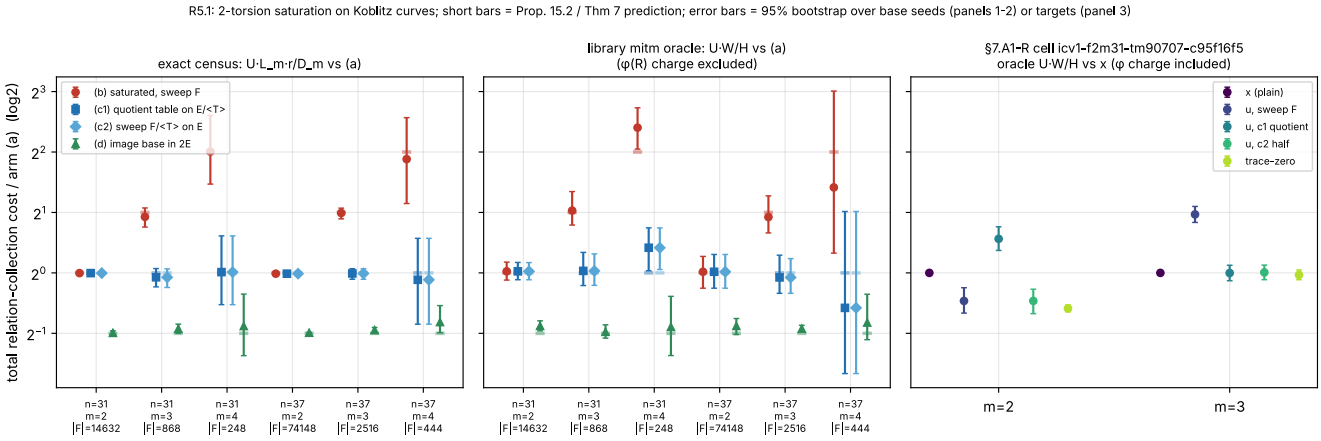


Figure 1: Recorded total relation-collection cost relative to the plain base (a), $\log_2$ scale. Short bars mark the Prop. 15.2 / Thm 7 predictions: (b) 2^{m-2}, (c) 1, (d) 1/2. Panels 1–2 show `icv1-f2m31-tm90707-c95f16f5` (6 base seeds) and `icv1-f2m37-tm534059-32aad96b` (5 seeds; 3 at $m = 4$), equal $|F|$ per cell. Panel 1 is the exact census over all of G. Panel 2 is the library `FrobeniusPairTable` / `decompose_mitm_frobenius` on 6000 shared targets (1500 at $m = 4$), with 95% bootstrap over seeds. Panel 3 is the §7.A1-R cell (divisor 0;1;2 bases), with paired bootstrap over 200 000 / 4000 targets. Sources: `summary.json`, `a1_m31.json`.

cell	law (b), (c), (d)	(b) census / oracle	(c1) census / oracle	(d) census / oracle
n=31, m=2	1, 1, 1/2	1.00 / 1.02	1.00 / 1.02	0.50 / 0.54
n=31, m=3	2, 1, 1/2	1.90 / 2.04	**0.95 / 1.02**	0.53 / 0.51
n=31, m=4	4, 1, 1/2	4.02 / 5.29	**1.01 / 1.33**	0.54 / 0.54
n=37, m=2	1, 1, 1/2	0.99 / 1.01	0.99 / 1.01	0.50 / 0.54
n=37, m=3	2, 1, 1/2	1.99 / 1.90	**0.99 / 0.95**	0.52 / 0.53
n=37, m=4	4, 1, 1/2	3.69 / 2.67	**0.92 / 0.67**	0.57 / 0.57

The (c1) oracle values exclude the φ(R) charge of one addition per target. With the charge, the $m = 2$ oracle value becomes 2.0, because the $m = 2$ probe is a single lookup; at $m \geq 3$ the charge changes the value by less than 0.2%. The 95% intervals are in `RESULTS.md`. At $m = 3$ every (c) interval contains 1 and excludes 2.

The pushforward identity $D_{m(\varphi(F_b))} = D_{m(F_b)}$ held exactly in 343/343 cells. In every cell, small n included, census (c1)/(b) is exactly 1/2 at $m = 3$ and 1/4 at $m = 4$. At $n \leq 23$ the bases hold only 1–4 Frobenius columns and their sums are structured τ-adic

combinations, not random. There the baseline (b)/(a) ranges over 0.5–7 and the class-factor law is not reached. Those cells are reported and not used for the verdict.

1.4. The §7.A1-R point (`icv1-f2m31-tm90707-c95f16f5`, $r = 1439393$)

- At $m = 3$ the u-arm sweeping F_u costs 1.96× the x-arm in total ([1.78, 2.14]) and 3.40× per relation. This reproduces the §7.A1-R loss, in a different unit.
- The same base with the quotient oracle costs **1.00× [0.91, 1.09]**. The half-sweep on E costs 1.01×. The trace-zero arm costs 0.98×.
- $\varphi(F_u)$ is exactly the trace-zero base translated by T: 589 of 589 abscissae are inverses of trace-zero abscissae. Hence D_3 = 318 618 for both.

1.5. Verdict

Confirmed. The falsifier is not met. In characteristic 2, an oracle on $F/\langle T\rangle$ removes the full 2^{m-2} kernel-saturation penalty. (c) ≈ (a) and (c) ≈ 2·(d) at $n = 31, 37$. The §7.A1-R u-frame loss disappears under the quotient oracle. Neither inseparability nor $x(T) = 0$ interferes: φ is the separable Verschiebung.

Limits:

- toy sizes;
- cost unit = counted additions + table lookups, with canonicalisations not priced and the table build not in the total;
- no linear algebra or full-rank streams;
- $m = 4$ oracle intervals wide;
- no algebraic quotient oracle.